

Fulton Starred problems

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This is a solution to some of the starred problems, as well as a few of the nonstarred ones. Rarely I will also give an alternative proof to a theorem. All problems will be solved using only other problems that I have written solutions for.

Chapter 1

1.1-1.7

These are skipped since I recall doing them in Atiyah Macdonald.

1.8

Note that k is a PID, therefore take a generator of the ideal I and note that $\mathbb{A}$ is just the roots of this polynomial. Hence finite.

1.9

Suppose $a = (a_1, \dots, a_n)$. The ideal

$$(X_1 - a_1, \dots, X_n - a_n)$$

vanishes only on a . Therefore take these ideals for all the desired points and multiply them together.

1.14

Write F as a polynomial in X_1

$$F = X_1^r f_r(X_2, \dots, X_n) + \dots + f_0(X_2, \dots, X_n)$$

by induction on n choose $X_2, \dots, X_n$ so that f_r is nonzero. Then using 1.4 choose X_1 so that the whole expression is nonzero. Therefore any nonzero polynomial has points it is nonzero on. Suppose FTSOC there are finitely many such points $a_1, \dots, a_k$, let b_i be the first index of these points. Then apply the lemma to

$$(X_1 - b_1) \cdots (X_k - b_k) F$$

a contradiction.

Again write F as a polynomial in X_1 . By the assumption there is at least one nonzero constant $f_k(X_2, \dots, X_n)$. There are infinitely many n -tuples $X_2, \dots, X_n$ which make f_k nonzero. Now by algebraically closed property we can choose X_1 so that the polynomial is 0. This gives infinite solutions.

1.15

Let I be the ideal generating V and J be the ideal generating W . Then take $I + J$.

1.16

If $V = W$ then it follows $\mathcal{I}(V) = \mathcal{I}(W)$. Suppose that $V \neq W$.

$$\begin{aligned}\mathcal{I}(V) &= \mathcal{I}(W) \\ \mathcal{I}(\mathcal{V}(\mathcal{I}(V))) &= \mathcal{I}(\mathcal{V}(\mathcal{I}(W))) \\ V &= W\end{aligned}$$

1.17

a

V algebraic means $V = \mathcal{V}(I)$ for some ideal. The condition that $P \notin V$ means that there is some $f \in I$ with $f(P) \neq 0$. Now take $f(X)/f(P)$.

b

Recall there is an ideal which vanishes on one point

$$a = (a_1, \dots, a_n) \quad J = (X_1 - a_1, \dots, X_n - a_n)$$

and we can union algebraic sets by intersecting ideals. Therefore apply (a) to the ideals formed by unioning V with all but one of the points.

c

Take:

$$\sum_j a_{ij} F_j$$

1.18

Expand using the binomial formula

$$(a + b)^{n+m} = \sum_{k=0}^{n+m} \binom{n+m}{k} a^k b^{n+m-k}$$

and observe that all the terms are in I . This shows that the Radical ideal is closed under addition. It is also closed under multiplication.

1.20

Note that $I \subseteq \text{Rad}(I)$ so that $\mathcal{V}(\text{Rad}(I)) \subseteq \mathcal{V}(I)$. Now note that zeros of a polynomial are invariant under taking powers. Therefore $\mathcal{V}(I) \subseteq \mathcal{V}(\text{Rad}(I))$.

1.21

Note that the quotient map is isomorphic to evaluation at $(a_1, \dots, a_n)$. This is a map into a field, hence the ideal is maximal. It also follows that the map $k \rightarrow k[X_1, \dots, X_n]/I$ is an isomorphism.

1.22

a

This is the lattice isomorphism theorem. Note that the preimage of an ideal under a ring homomorphism is an ideal and the preimage of 0 is I .

b

Note the characterizations:

(a) J is radical iff R/J is reduced.

(b) J is maximal iff R/J is a field.

(c) J is prime iff R/J is a domain.

The lattice theorem implies that $(R/I)/J' \cong R/J$ which gives all the desired results.

c

Take the images of a set of generators under the quotient map. This applies to all ideals so R/I is Noetherian if R is Noetherian.

1.29

In the case of infinite fields the polynomials which vanish everywhere $\mathcal{I}(\mathbb{A}^n(k))$ are only 0 (by 1.14). Then note that (0) is prime since polynomials form a domain.

1.37

Any higher power of X in the quotient can be reduced to one of these powers by the division algorithm.

1.38

Suppose that J is an ideal corresponding to an algebraic subset of V . Then $\sqrt{I+J}$ corresponds to the same algebraic subset by Nullstellensatz (and intersection property). Furthermore by Nullstellensatz distinct ideals of this form define distinct algebraic sets. Therefore there is a one-to-one correspondence between ideals of this type and algebraic subsets of V . Furthermore ideals of this type correspond uniquely to radical ideals in R/I by lattice theorem.

1.39

a

Suppose FTSOC that $\mathfrak{q} \subseteq \mathfrak{p} = (t)$ has the desired property. Then take $f \in \mathfrak{q}$ and factor it into irreducibles as:

$$f = t^n \cdot a_1 \cdots a_n$$

By primality of $\mathfrak{q}$ one of these terms must be contained in $\mathfrak{q}$. If it is t then $\mathfrak{p} \subseteq \mathfrak{q}$. If it is one of the a_i then $\mathfrak{q} \not\subseteq \mathfrak{p}$.

b

This is asking whether there is an ideal I so that $0 \subseteq I \subseteq (F)$. where F is prime. The result follows immediately from (a).

1.41

Choose the generators that generate S as a module, they also generate S as a ring.

1.43

Note that

$$K[v_1, \dots, v_n] \subseteq K(v_1, \dots, v_n)$$

because they are domains.

1.44

Multiply the denominators of all the generating elements together to get some c and then take $1/z$ where z is coprime to c . This element cannot be expressed in terms of the generators. This can be proven by taking an irreducible factor of z and inducing a valuation.

1.45

a

$$T + \sum Sw_j = \sum \sum Rv_i w_j$$

b

Use (a).

c

$$T = S(w_1, \dots, w_m) = R(v_1, \dots, v_n)(w_1, \dots, w_m) = R(v_1, \dots, v_n, w_1, \dots, w_m)$$

1.46

Given an element $t \in T$ write

$$t^n + a_{n-1}t^{n-1} + \dots + a_0 = 0 \quad a_i \in S$$

and note $R[a_0, \dots, a_{n-1}, t]$ is finite over R .

1.47

If S is integral over R then $R[s]$ is finite over R for all $s \in S$. Then take a set of ring generators s_i for S and note $S = R[s_1, \dots, s_n]$ is finite over R .

Suppose that S is finite over R . Then we have a set of module generators for $s_1, \dots, s_n$ for S . We need to show that every $s \in S$ satisfies a minimal polynomial in R . Write

$$s \cdot s_j = \sum r_{ij} s_i \quad r_i \in R$$

which gives a matrix $A_{ij} = r_{ij}$. Then $\det(A - s \text{Id})$ is a monic polynomial in s with coefficients in R that kills s (it acts by 0 in S).

1.48

a

This is the definition of an algebraically closed field.

b

This is by the universal property of algebraically closed fields.

1.49

a

Note that $K[X]$ is a UFD. Therefore a normal ring. This is by Gauss's lemma.

b

If there was such an element then $F^n z \in K[X]$ for sufficiently high n . Choose a G coprime to F and consider $1/G$.

1.50

a

Closure under addition and multiplication of a, b follows since if $R[a]$ and $R[b]$ are finite over R then $R[a, b]$ (and hence subfields of it) is also finite over R . Furthermore if we have x integral over R then:

$$x^n + a_{n-1}x^{n-1} + \cdots a_0 = 0 \implies x^{-n} + \frac{a_1}{a_0}x^{-(n-1)} + \cdots + \frac{1}{a_0} = 0$$

b

It can be shown that if an inverse to an element exists in an integral extension, then it exists in the base field. This is because.

$$x^n + a_{n-1}x^{n-1} + \cdots a_0 = 0 \implies x^{-1} = -\frac{1}{a_0}(x^{n-1} + a_{n-1}x^{n-2} + \cdots + a_1)$$

1.51

a

Multiplication by any nonzero element induces an injection of a finite dim K vector space. Hence invertable, the inverse can be found explicitly using Euclid's algorithm.

b

Fields maps are automatically injective. To show surjectivity note that the image of the map is $F[y]$. However since this is a field $F[y] = F(y)$.

c

G is killed by the isomorphism to $K(y)$, therefore it is contained in (F) .

d

In L , F can factor in this way since y is a root.

1.52

We can reduce the degree of the polynomial (split it) by considering in the field $K[X]/(f)$. Then use induction on all the factors it splits into.

1.53

Suppose that x is a repeated factor, then F and F' both kill x . Therefore by 1.51c F' divides F , a contradiction with irreducibility.

1.54

a

Take the degree n minimal polynomial for v and make the substitution $v = v'/a$ (where a is the lcm of the denominators). Then multiply through by a^n to clear denominators.

b

Choose a basis over K and then apply (a) to satisfy the condition.

Normal Basis Theorem

Let G be the Galois group of L/F with elements $\sigma_1 = e, \dots, \sigma_n$. Take a primitive element α with minimal polynomial f . Now write

$$g(x) = \frac{f(x)}{(x - \alpha)f'(x)} \quad g_i(x) = \sigma_i g(x) = \frac{b(x)f(x)}{(x - \sigma_i \alpha)f'(x)}$$

By considering the formula for the derivative (and noting that f is separable and therefore coprime to f') we may derive:

$$\begin{aligned} g_i g_j &\equiv 0 \pmod{f} & i &\neq j \\ g_1 + \dots + g_n &\equiv 1 \pmod{f} \\ g_i^2 &\equiv g_i \pmod{f} \end{aligned}$$

Now take the matrix $A_{ij} = (\sigma_i \sigma_j)g(\alpha)$. Note that each entry is 1 if $\sigma_i = \sigma_j^{-1}$, and otherwise 0. This is a permutation matrix therefore it has nonzero determinant. Then the conjugates of $g(\alpha)$ form a normal basis.

Chapter 2

2.1

There kernel are exactly those polynomials which vanish on V .

2.2

Pass to the quotient $\Gamma(V)$ and use lattice theorem. Then apply the Nullstellensatz in all cases (subvarieties of V correspond to radical ideals containing $\mathcal{I}(V)$).

2.3

a

The restriction is by passing to the quotient $\Gamma(V)/\mathcal{I}_V(W)$.

b

This is by the definition given in (a).

2.4

Using Corollary 4 it follows *i* and *ii* are equivalent since the functions on one point are all constant. It suffices to show that all finite varieties have a single point. This is by contrapositive, and because a single point always corresponds to a finite variety.

2.6

Draw the diagram

$$\begin{array}{ccccc} V & \xrightarrow{\phi} & W & \xrightarrow{\psi} & Z \\ & \searrow & \searrow & & \downarrow f \\ & & & & k \end{array}$$

and note $\widetilde{\psi \circ \phi} = (f \mapsto f \circ (\psi \circ \phi))$ and $\tilde{\phi} \circ \tilde{\psi} = (f \mapsto f \circ \psi \circ \phi)$ which are equal.

2.7

The variety $\phi^{-1}(X)$ is an algebraic subset. Take the ideal generated by the image of $\tilde{\phi}$. The way the book defines an algebraic set is alternatively just with a set of polynomials (rather than an ideal), that makes this argument easier.

Now suppose that $\phi^{-1}(X)$ is irreducible and X is contained in the image ϕ . Suppose FTSOC that $X = V \cup W$. Then we can take $\phi^{-1}(X) = \phi^{-1}(V) \cup \phi^{-1}(W)$. If W and V are nonzero and proper then so will their inverses.

2.9

a

Consider the image of a polynomial f vanishing on W' in W . We get $f \circ \phi$ where ϕ maps V' to W' . Therefore $f \circ \phi$ kills V' .

b

This is sort of tautological.

2.10

We have $T_i = x_i$ the projection onto just one index. A more general projection map will just be a sum of some of these.

2.14

a

The composition of a linear polynomial with a linear polynomial is a linear polynomial.

b

We need to have $\tilde{F}_i$ to be X_{m+i} . Project back to $\mathbb{A}^{n-1}$ by ignoring X_n and choose $T_1, \dots, T_{n-1}$. Then pull back, we have $\tilde{F}_i = X_{m+i} + a_i X_n$ and some others which are linearly independent in the full space but not in the projection. Postcompose with the map $X_{m+i} \mapsto X_{m+i} - a_i X_n$ so $\tilde{F}_i = X_{m+i}$. Then choose one of the extra polynomials $c + \sum_{i=0}^n c_i X_i$ and send $X_n \mapsto \frac{1}{c_n} \left(-c - \sum_{i=0}^{n-1} -c_i X_i \right)$. By construction all the other polynomials can be expressed as a linear combination of these.

Note that this construction is sort of similar to taking a matrix inverse using the Schur Complement. It's not exactly linear algebra though because of constant shift terms.

c

The m represents the maximal number of F_i which are linearly independent to each other. This property is agnostic to choice of basis, and any two maximal sets have the same size.

2.15

a

Affine changes of coordinates are linear transformations with a translation.

b

Get the ideal of P , that is (X_i, a_i) . Then forget the last equation and make the affine transformation taking $(a_1, \dots, a_{n-1}, a_n + 1)$ to Q but fixing P . This is by having

$$ae_n \mapsto (a - a_n)Q + (1 + a - a_n)P$$

c

This is by comparing dimension.

d

Move P to the center. Make the linear transformation that make the slopes match. Then translate P to P' .

2.21

This is by the universal property of localization. The image of every polynomial not 0 at P is invertable in the image, therefore the map extends uniquely.

2.22

a map being an isomorphism is preserved by localization at a prime ideal.

2.23

All the choices of uniformizers are related by a unit by the characterization of ideals.

2.24

a

Every polynomial is invertable except the factors of $X - a$.

b

If we take the reciprocal of everything we have exactly the DVR $\mathcal{O}_0(V)$.

2.26

a

This is the definition of a valuation ring. Otherwise let consider some $u\pi^a \in K$, where if a is negative the inverse in the DVR.

b

Everything not in the DVR is a unit. Therefore the maximal ideal of S cannot contain properly contain $\mathfrak{m}$.

2.28

Suppose that an element is not in the maximal ideal. Then it has valuation 0. We claim these elements are all units. Indeed, their inverse will also have valuation 0 and be contained in R . Now it suffices to show that the maximal ideal of R is principal. Take an element π with order 1. If b has valuation ≥ 1 then b/π has valuation ≥ 0 so that it is contained in R . Therefore $\mathfrak{m} = \pi R$.

Write every element as $u\pi^n$ where u is a unit and π is the uniformizer. Then take the order of $u\pi^n$ to be n . We can check that all the properties are satisfied.

2.29

Consider $a\pi^x + b\pi^y$ where $x < y$ and a, b are units. It suffices to show that $a + b\pi^{y-x}$ is a unit. This is because it is not contained in the Jacobson radical (those elements with $1 + xR \subseteq R^\times$) The Jacobson radical as the intersection of all maximal ideals is (π)

2.30

Let $z = z_0$ and recursively define:

$$z_{i+1} = z_i - t^i \lambda_i \quad z_i \equiv t^i \lambda_i \pmod{\mathfrak{m}^{i+1}}$$

2.35

a

This is a combinatorics problem. Using stars and bars there are in general

$$\binom{n+d-1}{d}$$

homogeneous polynomials of degree d in n variables.

b

Polynomials satisfy the conditions of a vector space. Furthermore any homogeneous polynomial can be written in terms of monomials and no nontrivial linear relation exists among distinct monomials.

c

There are in general $d+1$ distinct monomials of degree d in 2 variables. Therefore the vector space has dimension $d+1$ and it suffices to show that no nontrivial linear relation exists among the A_{ij} . We will use

the contrapositive and induction. Write $A_k = A_{k,d-k}$. Then note

$$\begin{aligned}\sum \lambda_k A_k &= 0 \\ \lambda_0 A_0 &= 0 \pmod{M_1} \\ \lambda_0 &= 0 \qquad L_1 \cdots L_d \text{ must be nonzero} \pmod{M_1} \text{ since it is a UFD}\end{aligned}$$

and similarly $\lambda_d = 0$ by considering $\pmod{L_1}$. Therefore we can factor out $L_1 M_1$ from the sum and reduce to the case of $d - 1$. In the base case $d = 1$ the theorem is clear.

2.38

This is by explicitly making basis.

2.39

This is tautological in commutative rings.

2.40

a

Note that

$$\sqrt{I^n + J^m} \supseteq \sqrt{I^n} + \sqrt{J^m} \supseteq I + J = R$$

b

It is a weaker assertion that the I_i are comaximal. Therefore this is a valid starting point. Take some $\lambda \in \bigcap I_k$. We need to express λ as contained in the product. Take $\alpha_{ij} + \beta_{ij} = 1$ for $\alpha_{ij} \in I_i$ and $\beta_{ij} \in I_j$ for all $j > i$. Then consider the product $\prod_{j>i} \alpha_{ij} + \beta_{ij}$ along all such expressions. By induction on N , each term in the expansion will be contained in one of the J_i (as defined in the problem). However it is equal to 1, therefore by multiplying by λ we see that λ is a product of terms in $\prod I_k$.

2.41

Suppose that $a_1, \dots, a_n$ are generators for I with $a_i^{k_i} \in I$. Then raise I to the power $\sum k_i$. Then by pigeonhole principle every possible expression in I^n will be contained in J .

2.42

a

This is the universal property of the quotient.

b

This is also by the universal property of the quotient.

2.43

This is because after localizing at $\mathfrak{p}$ then $\mathfrak{p}R_{\mathfrak{p}}$ is the unique maximal ideal of the localized ring $R_{\mathfrak{p}}$.

2.44

This is the third isomorphism theorem. In particular we have

$$\begin{aligned}
 \frac{\mathcal{O}_P(V)}{J'\mathcal{O}_P(V)} &= \frac{\Gamma(V)}{J'} && \text{exactness of localization} \\
 &= \frac{\Gamma(\mathbb{A}^n)/\mathcal{I}(V)}{J\Gamma(\mathbb{A}^n)/\mathcal{I}(V)} \\
 &= \frac{\Gamma(\mathbb{A}^n)}{J\Gamma(\mathbb{A}^n)} && \text{third isomorphism theorem} \\
 &\cong \frac{\mathcal{O}_P(\mathbb{A}^n)}{J\mathcal{O}_P(\mathbb{A}^n)} && \text{exactness of localization}
 \end{aligned}$$

2.45

Use contrapositive, then there is some point x in $\mathcal{V}(I) \cap \mathcal{V}(J)$. Therefore $I + J$ is contained in $\mathfrak{m}_x$. Therefore they are not comaximal.

On the other hand if $I + J$ is not the whole ring then it is contained in a maximal ideal $\mathfrak{m}_x$. Then x will be contained in $\mathcal{V}(I) + \mathcal{V}(J)$.

2.46

The dimension in the number of forms of degree less than some number. Then apply the results of 2.35.

Proposition 6

I strongly dislike the given proof of proposition 6. Therefore here is a proof using commutative algebra. Consider a primary decomposition of I . We claim that the prime ideals appearing in the primary decomposition are exactly the $\mathfrak{m}_{P_i}$. This is because the radical of I is exactly $\bigcap \mathfrak{m}_{P_i}$ by the Nullstellensatz. Therefore the $\mathfrak{m}_{P_i}$ are minimal prime ideals containing I by Nullstellensatz. Now write

$$I = \bigcap \mathfrak{q}_i \quad \mathfrak{q}_i \text{ is } \mathfrak{m}_{P_i} \text{ primary}$$

and note that by the Noetherian condition there are n_i so that $\mathfrak{m}_i^{n_i} \subseteq \mathfrak{q}_i$. Therefore the $\mathfrak{q}_i$ are comaximal and we can use CRT to write

$$I \cong \prod_{\mathfrak{q}_i} \frac{k[X_1, \dots, X_n]}{\mathfrak{q}_i}$$

however since the $\mathfrak{q}_i$ are $\mathfrak{m}_i$ primary (and these are maximal ideals), each one is only contained by $\mathfrak{m}_i$. Therefore these are all local rings and we can localize at the $\mathfrak{m}_i$ without effecting the summands. This yields

$$I \cong \prod \frac{\mathcal{O}_i}{\mathfrak{q}_i \mathcal{O}_i} = \prod \frac{\mathcal{O}_i}{I \mathcal{O}_i}$$

where we have replaced

$$I \mathcal{O}_i = \bigcap_j \mathfrak{q}_j \mathcal{O}_i = \mathfrak{q}_i \mathcal{O}_i$$

by the same reasoning as above (ideals which are not contained in $\mathfrak{m}_i$ become the whole ring in the localization).

2.47

Note that R is finite type over k so we may write it as $k[X_1, \dots, X_n]/I$. Furthermore since R is finite dimensional over k we know that I has finite locus by a corollary of Nullstellensatz. Therefore apply Proposition 6.

2.48

This is the first isomorphism theorem.

2.49

a

This is the universal property of quotient/cokernel.

b

This is the third isomorphism theorem. It can be proven using the universal property of the quotient.

c

Apply the exact sequence in (b) and note that the sequence splits. Therefore the vector space is isomorphic to a direct sum and the dimensions add.

d

Take the exact sequence

$$0 \rightarrow I \rightarrow R \rightarrow R/I \rightarrow 0$$

and quotient everything by J .

e

Note that $\mathfrak{m}^n$ is an ideal of $\mathcal{O}/\mathfrak{m}^{n+1}$. Then apply the cokernel exact sequence on the quotient map.

2.50

a

Note that $\mathfrak{m}$ is principal. Therefore $\mathfrak{m}^n/\mathfrak{m}^{n+1}$ is generated by a single element.

b

We have the exact sequences of k vector spaces

$$0 \rightarrow \mathfrak{m}^n/\mathfrak{m}^{n+1} \rightarrow R/\mathfrak{m}^{n+1} \rightarrow R/\mathfrak{m}^n \rightarrow 0$$

which proves $\dim_k R/\mathfrak{m}^{n+1} = \dim R/\mathfrak{m}^n + 1$. Induct on this to get the desired result.

c

This is by definition of ord and (b).

2.52

This is the diamond isomorphism theorem.

2.53

a

Note that the isomorphism can be explicitly constructed as

$$a + W \mapsto T(a) + T(W)$$

b

This is diamond isomorphism theorem applied twice.

c

Note that

$$\begin{aligned}
\frac{V}{W} &= \frac{V}{W+T(V)} + \frac{W+T(V)}{W} \\
&= \frac{V}{W+T(V)} + \frac{T(V)}{W \cap T(V)} \\
&= \frac{V}{W+T(V)} + \frac{T(V)}{T(W)} - \frac{W \cap T(V)}{T(W)} \\
0 &= \frac{V}{W+T(V)} - \frac{W \cap T(V)}{T(W)}
\end{aligned}$$

as desired (the dims in front of each term are implied).

d

By the previous part and 2.49(c) it suffices to show that

$$\dim \frac{W+T(V)}{(V)} = \dim \frac{W}{W \cap T(V)}$$

which is by diamond isomorphism theorem.

Chapter 3

3.10

Note that multiple points are common roots of F, F_X, F_Y . We assume WLOG that F_X is nonzero and consider the decomposition of $\mathcal{V}(F) \cap \mathcal{V}(F') =$

3.12

We claim that the degree of the intersection is exactly the lowest power of X appearing in the expression for F . We may write:

$$F = Y + YF_1 + X^n F_2 \quad F_1 = aX + bY + \dots \quad F_2 = k + \dots$$

We now wish to consider the order of Y in the quotient field $\frac{k[X,Y]_{\mathfrak{m}_0}}{(F)}$. We can write:

$$Y \cong -YF_1 - X^n F_2 \pmod{(F)}$$

Note that the order of $-YF_1$ is necessarily larger than the order of Y due to our expression for F_1 . Therefore by the properties of DVRs:

$$\text{ord } Y = \text{ord } X^n F_2 = \text{ord } X^n$$

since F is invertible in the DVR (it has nonzero constant term). The order of X is 1 since it intersects transversally, therefore the order of Y is n .

3.13

Note in the case that $n < m_P(F)$ we know that (F) is contained in $\mathfrak{m}^n$. Therefore we may write

$$\dim \frac{\mathfrak{m}^n}{\mathfrak{m}^{n+1}} = \dim \frac{(X, Y)^n \mathcal{O}_P(\mathbb{A}^2)/(F)}{(X, Y)^{n+1} \mathcal{O}_P(\mathbb{A}^2)/(F)} = \dim \frac{(X, Y)^n \mathcal{O}_P(\mathbb{A}^2)}{(X, Y)^{n+1} \mathcal{O}_P(\mathbb{A}^2)} = \dim \frac{(X, Y)^n}{(X, Y)^{n+1}}$$

by third isomorphism theorem. This has dimension $n + 1$ by known results.

3.19

This is a direct application of property 5.

3.21

Via a change of coordinates we can send this line to Y . Then the points of $F \cap Y$ are exactly the solutions to F when $Y = 0$. Furthermore all the intersection are transversal and therefore by the properties the sum of the intersection numbers is the X degree of F . This is equivalent to factoring F into the desired form.

3.24

a

Note that the spaces have the same dimension by previous results. Therefore it suffices to show that the map is injective. Note that $\bar{X}$ and $\bar{Y}$ must generate $\mathfrak{m}/\mathfrak{m}^2$ as a $R/\mathfrak{m} \cong k$ vector space since they generate $\mathfrak{m}$. Therefore $\bar{X}$ and $\bar{Y}$ are linearly independent.

b

The inequality $I(P, F \cap L_i) > m$ is due to property 5. Note that $L_i \neq \lambda L_j$ since this is an ordinary multiple point. Therefore any two L_i, L_j generate the forms of degree 1 in $k[X, Y]$. Since these generate $\mathfrak{m}/\mathfrak{m}^2$ by (a) and a dimensionality argument $\bar{l}_i$ and $\bar{l}_j$ must be linearly independent.

c

Note that by property 5 that each G_i shares a tangent line with F . Furthermore by the condition that $\bar{g}_i \neq \lambda \bar{g}_j$ that the G_i have an ordinary point at 0 (since the images otherwise are 0) and furthermore that their images are $\bar{l}_i$ where $\bar{l}_i$ is the image of the tangent that is shared. Therefore the shared tangents L_i are pairwise distinct. Therefore we have at least m distinct tangent lines and therefore P is an ordinary point.

d

We have proved the forward direction and the converse in the previous parts.

Chapter 4

4.2

Choose a set of coordinates. Note that we can write

$$f(\lambda) = F(\lambda x_1, \dots, \lambda x_n) = \sum \lambda^i F_i(x_1, \dots, x_n)$$

as a polynomial in λ . By the condition that $f(\lambda)$ is identically 0 this polynomial must be 0 and each $F_i(x_1, \dots, x_n) = 0$.

4.4

The forward direction is clear from the definition of a prime ideal. In the backwards direction, suppose that $AB \in I$. Write $A = \sum A_i$ and $B = \sum B_i$ as a sum of their homogeneous components. We have

$$(AB)_n = \sum_{i+j=n} A_i B_j = 0 \pmod{I}$$

and note that each of these terms must be in the ideal. If we choose n to be maximal then this implies $A_{\deg A} \in I$ or $B_{\deg B} \in I$. Suppose WLOG that $A_{\deg A} \in I$. Then we may ignore the highest order homogeneous component of A . The claim follows from induction on $\deg A + \deg B$.

4.6

Note that under the definition of $\mathcal{V}_P$ the properties (1), (2), (3), (4), (6), (8), (9), and (10) are all true with no modification. We have to modify (5) and (7).

5

Suppose WLOG that the point is represented by $(a_1, \dots, a_n, 1)$ (since atleast one index is nonzero). Then the maximal ideal of this point is $(X_1 - a_1 X_{n+1}, \dots, X_n - a_n X_{n+1})$.

7

The first two assertions still true. We replace the ideal in the third assertion by the one described in the correction to 5.

4.7

Let V be a cone. A irreducible component of V corresponds to a maximal irreducible subset. This is equivalent to a minimal prime ideal $\mathfrak{p}$ containing $\mathcal{I}(V)$. Take a finite set of homogeneous generators for $\mathcal{I}(V)$. Note that the ideal $\tilde{\mathfrak{p}}$ generated by the homogeneous elements of $\mathfrak{p}$ contains these generators and therefore contains $\mathcal{I}(V)$. Furthermore note by 4.4 that $\tilde{\mathfrak{p}}$ is also a prime ideal. By minimality $\tilde{\mathfrak{p}} = \mathfrak{p}$ and furthermore $\mathfrak{p}$ is homogeneous since it is generated by homogeneous elements.

4.9

Lift out of the quotient. By previous results forms of some degree form a finite dimensional vector space over k . Take a basis and project back into the quotient.

4.11

Note that a projective change of coordinates in $\mathbb{P}^n$. Note that we may consider this problem in $\mathbb{A}^{n+1}$, proving all the claims by 2.14. Indeed, this just becomes a special case of 2.14 in the case that the change of coordinates and the ideal are homogeneous. Then go back to $\mathbb{P}^n$.

4.12

Lift into $\mathbb{A}^{n+1}$. By 2.14 (or its proof) the dimension of a linear subvariety is n minus the minimal number of polynomials which generate it. Since $m < n + 1$ the dimension of this intersection of hyperplanes is at least 1 (a line) and furthermore by the projective assumptions intersects the origin. Therefore corresponding to atleast one point in $\mathbb{P}^n$.

4.13

a

If we lift up to $\mathbb{A}^{n+1}$ then a line in projective spaces uniquely corresponds to a plane passing through the origin in affine space. The assertion becomes equivalent to the statement that under a linear transformation the image of a plane is determined by the image of the two lines defining it. Then apply 2.15a.

b

In $\mathbb{P}^2$ we can define a line passing through (a, b, c) and (a', b', c') .

c

In $\mathbb{P}^2$ a line is a plane in $\mathbb{A}^3$, which has codimension 1 as desired.

d

We claim that changes of coordinates on $\mathbb{P}^2$ act double transitively on lines. By 4.14 we can send pairs of points generating the lines in $\mathbb{P}^2$ to each other. This will send any two lines to any other two lines.

4.14

This result can actually be strengthened. Take $n + 2$ points in $\mathbb{P}^n$ such that any $n + 1$ of them generate the entire projective plane. This is called the points being in projective general position. This translates to $n + 2$ vectors $v_1, \dots, v_{n+2}$ (up to scaling) in $\mathbb{A}^{n+1}$ such that there is only one nontrivial linear relation among any subset of $n + 1$. We claim that we can map these to any other $n + 2$ points. Take representatives $w_1, \dots, w_{n+2} \in \mathbb{A}^{n+1}$, we desire a map $T : v_i \mapsto w_i$. By reordering the v_i we may assume WLOG that w_{n+2} can be written in terms of $w_1, \dots, w_{n+1}$. By making a change of coordinates we can assume that the $v_1, \dots, v_{n+1}$ are the basis vectors for $\mathbb{A}^{n+1}$. Therefore take a matrix with columns $T = [k_1 v_1, \dots, k_{n+1} v_{n+1}]$ scalar multiples of the v_i . This almost sends each v_i to a scalar multiple of w_i , however we must consider:

$$v_{n+2} = a_1 v_1 + \dots + a_{n+1} v_{n+1} \quad [k_1 v_1, \dots, k_{n+1} v_{n+1}] v_{n+2} = k_1 a_1 v_1 + \dots + k_{n+1} a_{n+1}$$

By the condition we enforced all the a_i must be nonzero (otherwise we have a linearly dependant subset of $n + 1$) so we can choose the k_i so that this sum is w_{n+2} . Now note that T maps all the v_i to scalar multiples of the w_i . Furthermore note that T has been determined up to one minus the nullity of the matrix with columns w_i (since scaling T doesn't change its action on $\mathbb{P}^n$).

Note that we can much more easily prove that projective changes of coordinates act $n + 1$ transitively without the assumption of projective general position.

4.15

Any two distinct planes in $\mathbb{A}^3$ intersect at a line.

4.16

Map the 3 distinct intersection points to each other by 4.14 and the lines are determined.

4.17

Write $z = a/b$ as a rational function in lowest terms. Then the pole set is the projective variety generated by the ideal (b) .

4.19

By definition this ideal contains F^* . Therefore we must show that if $G \in (F)^*$ then $G \in (F^*)$. The former implies we can write G as a product

$$G = \sum A_i (B_i F)^* = \sum (A_i B_i^*) F^*$$

as desired.

4.20

a

Note that if we quotient out by this ideal (setting $Y = X^2$ and $Z = X^3$) we just get $k[X]$. This is a domain and therefore the ideal is prime hence radical and $\mathcal{I}(V) = (Y - X^2, Z - X^3)$.

b

Note that

$$Z - XY \equiv X^3 - X \cdot X^2 = 0 \pmod{(Y - X^2, Z - X^3)}$$

Now homogenize both sides to yield that:

$$ZW - XY \in \mathcal{I}(V)^*$$

Now we need to show that

$$ZW - XY \notin (YW - X^2, ZW^2 - X^3)$$

Indeed, suppose FTSOC that we have an expression of the form

$$f(YW - X^2) + g(ZW^2 - X^3) = ZW - XY$$

however we can try to solve this in the quotient of (X, Y) . This yields

$$\bar{g}ZW^2 = ZW$$

which clearly cannot be solved since there is no inverse to W .

4.21

This problem should be referencing 1.39. It follows from 1.39 after a reduction to the affine variety $\mathbb{A}^{n+1}$.

4.22

Suppose that $H_\infty \subseteq V$. We can dehomogenize to yield $0 \subseteq V_*$ so $V_* = 0$. Homogenizing again yields $H_\infty = V_*$ or $0 = V_*$ since both 0 and H_∞ dehomogenize to 0.

4.23

Note that by the previous parts every subvariety in $\mathbb{P}^n$ is one of

- (a) The empty set
- (b) The whole space $\mathbb{P}^n$.
- (c) One of the H_∞ .
- (d) The projective closure of some irreducible variety in $\mathbb{A}^n$.

the first 3 are very easily classified, so we focus on the fourth. We note that a variety can be obtained as the projective closure of a variety in $\mathbb{A}^n$ by means of φ_i iff it intersects U_i . This is because intersecting U_i is equivalent to not being contained in H_∞ . Then note it cannot contain H_∞ by 4.22 (unless it is the whole space) and apply property 7 of proposition 3. In particular, since every variety intersects one of the U_i all can be obtained in this way.

In the case of $\mathbb{P}^1$ this corresponds to single points. In the case of $\mathbb{P}^2$ this corresponds to the projective closure of irreducible curves and points, corresponding bijectively to homogeneous irreducible forms.

4.24

Note that P is contained in U_2 . Therefore by 4.23 a line through P corresponds to either L_∞ or the projective closure of a line in $\mathbb{A}^2$ that intersects 0. These are lines of the form $X - \lambda Z$ up to scaling.

4.25

a

It is a hyperplane because it is defined by a single linear polynomial.

b

For any finite set of points in $\mathbb{P}^n$ over any infinite field, there is a hyperplane not passing through any of them. Let $V = \mathbb{A}^{n+1}$. Consider finitely many points $v_1, \dots, v_n \in \mathbb{P}^n$. The linear polynomials correspond to elements of V^* (dual space) and the v_i corresponds to hyperplanes in V^* defined by $f(v_i) = 0$. Therefore it suffices to show that no vector space is a union of its proper subspaces.

Suppose FTSOC that $V = V_1 \cup \dots \cup V_n$ and that we have ignored all redundant terms in the union. Therefore we may assume that V_n and $V_1 \cup \dots \cup V_{n-1}$ do not include each other. Take elements $x \in V_n \setminus (V_1 \cup \dots \cup V_{n-1})$ and $y \notin V_n$. Now consider the solutions to:

$$x \equiv ty \pmod{V_i}$$

For the case of V_n we only have $t = 0$. Furthermore for $i \neq n$ we have $x \not\equiv 0 \pmod{V_i}$ so t has at most 1 solution. Since the base field is infinite can choose t avoiding all these solutions and obtain an element $x - ty$ not contained in $V_1 \cup \dots \cup V_n$.

4.26

a

The maps are just the compositions of natural maps

$$\mathbb{A}^{n+m} \cong \mathbb{A}^n \oplus \mathbb{A}^m \rightarrow U_i \times U_j \subseteq \mathbb{P}^n \times \mathbb{P}^m$$

We can define the biprojective closure as the smallest biprojective set containing the image of an algebraic set under $\varphi_{i,j}$. Given some polynomial $f \in k[X_1, \dots, X_n, Y_1, \dots, Y_r]$ we first homogenize wrt $X_1, \dots, X_{n+1}$ and then wrt $Y_1, \dots, Y_{n+1}$ in the usual way.

We can redefine the following terms in the context of $\mathbb{P}^{n_1} \times \dots \times \mathbb{P}^{n_r}$.

- (a) $-^*$ should mean homogenizing separately wrt each collection of variables
- (b) $-_*$ should mean dehomogenizing wrt to each collection.
- (c) H_∞ will be $(\mathbb{P}^{n_1} \times \dots \times \mathbb{P}^{n_r}) \setminus (U_1 \times \dots \times U_r)$.

so that all the statements of proposition 3 are still true as written. The proofs are very nearly the same.

b

This is by the same reasoning as 4.22 and 4.23, which establishes the desired claim and can be generalized to this multiprojective case. To show that the function fields are isomorphic, note the isomorphism

$$\alpha : k(V^*) \rightarrow k(V) \quad \alpha(f/g) = f_*/g_*$$

(it is somewhat simple to check injectivity and surjectivity). This restricts to an isomorphism:

$$\mathcal{O}_P(V^*) \rightarrow \mathcal{O}_P(V)$$

4.27

Take denominators and apply the same reasoning as the simple case.

4.28

a

One can see that scaling x or y scales the image by the same factor, therefore the map is well defined. Furthermore the map is injective since for the image to be 0 all the x_i or all the y_i have to be 0. This is disallowed by the definition of the projective plane.

b

The map induces a pullback on the polynomials defined by

$$T_{ij} \mapsto X_i Y_j$$

which one can verify is indeed a pullback.

c

We can see that V contains the image of S by direct computation, therefore it remains to show that every element of V is in the image of S . Define

$$X_i = \frac{T_{i0}}{T_{00}} \quad Y_j = T_{0j}$$

Now in the quotient we use the relation with $j = l = 0$.

$$T_{ij}T_{00} = T_{i0}T_{0j} \quad \implies \quad T_{ij} = X_i Y_j$$

as desired.

d

This is by parts (a) – (c) and problem 2.7.

Chapter 5

5.1

Note that the following solution was mainly a point of curiosity to see whether this could be proven without making a convenient affine change of coordinates. The commutative algebra is interesting but this is not the easiest/shortest proof by a long shot. That proof might be changing P to $[0 : 0 : 1]$ and using (6) of section

1.1 (my friends solution).

We assume WLOG that $P \in U_3$ and write $P = [a : b : 1]$. This is a multiple point iff $F_* \in I^2$ where $I = (X - a, Y - b)$. We claim that:

$$F = Z^n(F_*)^* \in (I^2)^* = (I^*)^2$$

By the exposition of the projective closure, I^* is the unique largest homogeneous ideal contained in $(X - a, Y - b, Z - 1)$. Therefore this condition is equivalent to $F \in (X - a, Y - b, Z - 1)^2$ (since F is always homogeneous). Finally by the results of section 2 this is equivalent to:

$$F_X(P) = 0 \quad F_Y(P) = 0 \quad F_Z(P) = 0$$

We still need to show that $(I^2)^* = (I^*)^2$. It is always true that $(I^*)^2 \subseteq (I^2)^*$ since $(fg)^* = f^*g^*$. Now consider a generic element $h \in (I^2)^*$:

$$h = \sum_i a_i \left(\sum_j f_{ij} g_{ij} \right)^* \quad f_{ij}, g_{ij} \in I$$

We claim that $Z^n h \in (I^*)^2$ for sufficient n . It suffices to show for each of the inner sums

$$\sum_j f_{ij}^* g_{ij}^* = Z^n \left(\left(\sum_j f_{ij}^* g_{ij}^* \right)^* \right)^* = \left(\sum_j f_{ij} g_{ij} \right)^*$$

however the LHS is an element of $(I^*)^2$ as desired. However $k[X, Y, Z]$ is a UFD, so $(I^*)^2$ is a primary ideal since it is a power of a prime ideal. Therefore either $h \in (I^*)^2$ or $Z^n \in \sqrt{(I^*)^2} = I^*$ which is not true since $Z \notin I$. Therefore $h \in (I^*)^2$.

Note that this argument can more easily generalize to the statement of problem 3.5 in the projective case.

5.4

Recall in section 4.3 the isomorphism $\mathcal{O}_P(\mathbb{A}^n)$ with $\mathcal{O}_P(\mathbb{P}^n)$ defined by

$$\frac{f}{g} \mapsto \frac{f_*}{g_*}$$

I claim an isomorphism:

$$\mathcal{O}_P(\mathbb{A}^2/(F, G)) \rightarrow \mathcal{O}_P(\mathbb{P}^2/(F/L_1^{d_1}, G/L_2^{d_2}))$$

By the isomorphism of $\mathcal{O}_P(\mathbb{A}^2)$ and $\mathcal{O}_P(\mathbb{P}^2)$ it suffices to prove that the image of $(F/L_1^{d_1}, G/L_2^{d_2})$ is exactly (F_*, G_*) . This is by the explicit map and because L_1 and L_2 are invertable in the image.

Now suppose WLOG that $P \in U_3$ so that $P = [a : b : 1]$. By the isomorphism of the local rings, the tangent line to F as P is the projective closure of the tangent line to F_* at $(0, 0)$. This is the line

$$F_{*X}(P)(X - a) + F_{*Y}(P)(Y - b) = 0$$

which lifts to:

$$F_X(P)X + F_Y(P)Y + (-aF_X(P) - bF_Y(P))Z = 0$$

Note from $XF_X + YF_Y + ZF_Z = mF$ we can write $-aF_X(P) - bF_Y(P) = mF(P) + 1 \cdot F_Z(P) = F_Z(P)$ and we obtain the line

$$F_X(P)X + F_Y(P)Y + F_Z(P)Z = 0$$

5.5

When we dehomogenize wrt Y we can compute the multiplicity in $\mathbb{A}^2$ using those methods. That yields that $m_P(F)$ is the smallest m so that $F_m \neq 0$. Furthermore by our proof of 5.4 the tangents are the projective closure of the tangents in $\mathbb{A}^2$. Therefore since the tangents in $\mathbb{A}^2$ are determined by the factorization of F_m we are done.

5.6

We will prove the claim for differentiating wrt X , Y or Z . Due to the symmetrical nature of the claim, we may assume that $P \in U_3$ and $P = [a : b : 1]$. Therefore take F_* . The multiplicity is the order of the lowest order homogeneous component. If we differentiate wrt X or Y the order will necessarily go down. Differentiating wrt Z before dehomogenizing will not change the order of the lowest order term by writing

$$F = \sum F_i(X, Y)Z^{n-i} \quad \deg F_i = i$$

and explicitly computing the terms.

5.7

Consider an intersection point P and assume WLOG $P \in U_3$. Then we can take F_* and G_* in $\mathbb{A}^2$ since they are not contained in or containing H_∞ . Now the result follows from proposition 2 of section 1.6.

5.8

a

If $F_X = F_Y = F_Z = 0$ then F is constant, therefore either nothing or the entire projective line or nothing.

b

Note that at a multiple point F and F_X intersect. Therefore since F is irreducible (has no common factor with its derivative) the results follows by 5.7.

5.12

This is the analogue of problem 3.21. The reduction to the affine case is by the assumption on the point.

5.14

Pass to $\mathbb{A}^3$ and make a linear algebra argument. We need to show that there are infinite planes passing through exactly one line in a collection. This is because of linear algebra facts. In particular, pass to the quotient space $\mathbb{A}^3/L_1$. This has dimension 2 and is therefore infinite, so we may choose a point not equivalent to any of the other lines.

5.15

Consider the order of one of the lines L_i we construct in 5.14. By passing to the affine space we can see that each of these lines has order 1 on P_i and order 0 on $P_{j \neq i}$. When we multiply these lines the orders add, therefore take the product

$$z = \prod L_i^{m_i} L_0^{-\sum m_i}$$

where L_0 is a line not passing through any of these points, having order 0 on every one.

5.16

Suppose that $P = [a : b : 0]$ and $F(P) = 0$. Since F is homogeneous then $aF_X(P) + bF_Y(P) = 0$. Suppose FTSOC that $F_X(P) = 0$. Then it is clear that $F_Y(P) = 0$. The tangent line to F at P is then

$$F_Z(P)Z = 0$$

however Z is not tangent to P by the intersection number, a contradiction.

Theorem 1

1

Recall that the curves of degree d such that $m_{P_i}(F) \geq r$ form a linear subvariety of codimension $\frac{r_i(r_i+1)}{2}$. By induction, it suffices to prove the lemma that if $A, B \subseteq V$ are vector subspaces of codimension n and m then $V \cap W$ has codimension at most $n + m$. Consider the natural map

$$\phi : A \oplus B \rightarrow V \quad \phi(a \oplus b) = a + b$$

Let r be the dimension of V and h be the dimension of the kernel. By rank nullity theorem we have

$$2r - n - m - h = \dim(\text{im } \phi) \leq r$$

$$h \geq r - n - m$$

with equality iff $A + B = V$. However, the kernel is in 1 to 1 correspondance with the intersection. Therefore the codimension of $A \cap B$ is at most $n + m$.

2

Call V_i the subspace of curves of degree d such that $m_{P_i}(F) \geq r_i$. By the above proof and induction, in order for $\text{codim} \bigcap V_i = \sum \frac{r_i(r_i+1)}{2}$ we need

$$V_{k+1} + \bigcap_{i=1}^k V_i = V$$

since this is the condition to have equality of dimension at each step. Intuitively we must show that the V_i are "maximally disjoint". It suffices to manifest $\text{codim } V_{k+1} = \frac{r_{k+1}(r_{k+1}+1)}{2}$ elements of $\bigcap_{i=1}^k V_i$ which are linearly independent in the quotient by V_{k+1} . Choose lines L_i passing through P_i but not $P_{j \neq i}$. Then take

$$z = F \prod_{i=1}^k L_i^{r_i} \quad \deg F = d - \sum_{i=1}^k r_i$$

we assume WLOG that $P_{k+1} = [0 : 0 : 1]$, then V_{k+1} consists of all polynomial whose minimal term has X, Y degree $> r_{k+1}$. Therefore our $\frac{r_{k+1}(r_{k+1}+1)}{2}$ linearly independent choices are represented by

$$X_a Y_b Z^{\deg F - a - b} \quad a + b < r_{k+1}$$

which we can do since $\deg F$ is necessarily larger than r_{k+1} by the condition. There are $\frac{r_{k+1}(r_{k+1}+1)}{2}$ such choices and we are done.

5.17

First suppose that all the points lie on a line. Then by problem 5.12 any conic intersecting all of them will have a common factor with the line. Therefore the conic has the form

$$FL \quad \deg F = 1$$

where L is the line. Our choices for F are spanned by X, Y , giving two dimensions.

Now suppose FTSOC that the points do not all lie on a line. Therefore make a projective change of coordinates to send P_1, P_2, P_3 to $[1 : 0 : 0], [0 : 1 : 0], [0 : 0 : 1]$. Then the space of possible polynomials is spanned by XY, XZ , and YZ . Therefore the dimension of these polynomials in the projective space is 2. Now enforcing the constraint of being 0 on another point reduces the dimension by 1. This is because not all of these can be 0 on that point unless the point is one of the P_i .

5.18

The fact that there is only one conic follows from 5.18.

5.19

By the proof of the previous part any such conic can be written as $aXY + bXZ + cYZ$. Then the constraint of being 0 on these 2 points turns out to give nontrivial linear relations which determine a, b, c up to a constant.

5.19

Take polynomials of the form:

$$tX(X-1)(X-2) + (1-t)Y(Y-1)(Y-2)$$

for $t \in [0, 1]$.

Bezout's Theorem

step 1

The given proof of step 1 is good.

step 2

The given proof of step 2 is good.

step 3

I will give an alternate proof that $\dim \Gamma_* = \dim \Gamma_d$ for sufficiently large d . Construct the commuting diagram

$$\begin{array}{ccccc} \Gamma_1 & \xrightarrow{Z} & \Gamma_2 & \xrightarrow{Z} & \Gamma_3 & \xrightarrow{Z} & \cdots \\ \downarrow \phi_1 & \swarrow \phi_2 & & \searrow \phi_3 & & & \\ \Gamma_* & & & & & & \end{array}$$

where the top maps are multiplication by Z and the bottom maps are dehomogenization. We first want to prove that the dehomogenization is an injective map. Write:

$$\begin{aligned} f \mapsto 0 &\implies f_* = AF_* + BG_* \\ &\implies (f_*)^* = (AF_* + BG_*)^* \\ &\implies f = Z^k (AF_* + BG_*)^* \end{aligned}$$

Now we can write

$$AF + BG = Z^r ((AF + BG)_*)^* = Z^r (AF_* + BG_*)^*$$

so that:

$$f = Z^{k-r} (AF - BG)$$

at which point it follows that $f = A'F + B'G$ because multiplication by Z is injective in Γ . Furthermore, given an element $f \in \Gamma_*$ we can homogenize it so that it is the image of some $f^* \in \Gamma_d$. Therefore the image of all of these maps is Γ_* . We also note that $\Gamma_d \xrightarrow{Z} \Gamma_{d+1}$ is an isomorphism for all sufficiently large d since the maps are injective and the spaces have the same dimension by step 1. Therefore the ϕ_d are isomorphisms for sufficiently large d since they are injective and surjective.

5.21

We use contrapositive. If F is not reducible then write $F = F_1 \cdot F_2$. By Bezout's theorem F_1 and F_2 will have an intersection point where F is singular.

5.22

Since F is irreducible it shares no common component with F_X , therefore write

$$\sum m_P(m_P(F) - 1) \leq \sum m_P(F)m_P(F_X) \leq n(n-1)$$

where the first inequality is by problem 5.6 and the second is by Bezout's theorem.

5.23

a

We have $F^T = F \circ T$. Therefore by multivariable chain rule

$$F_i^T = [(\nabla F) \circ T](D_i T)$$

where D_i denotes the total derivative wrt $i \in \{X, Y, Z\}$. More explicitly

$$F_i^T = \sum_{a=X,Y,Z} \left(\frac{\partial T^a}{\partial i} \right) \cdot \left(\frac{\partial F}{\partial a} \circ T \right)$$

where T^a denotes the a component of T . Now note that T is linear so when we differentiate wrt $j \in \{X, Y, Z\}$ and apply the product rule we just get

$$\begin{aligned}
F_{ij}^T &= \frac{\partial}{\partial j} \sum_{a=X,Y,Z} \left(\frac{\partial T^a}{\partial i} \right) \left(\frac{\partial F}{\partial a} \circ T \right) \\
&= \sum_a \left(\frac{\partial T^a}{\partial i} \right) \frac{\partial}{\partial j} \left(\frac{\partial F}{\partial a} \circ T \right) \\
&= \sum_a \sum_b \left(\frac{\partial T^a}{\partial i} \right) \left(\frac{\partial T^b}{\partial j} \right) \left(\frac{\partial^2 F}{\partial a \partial b} \circ T \right) \\
&= \sum_a \sum_b T_{ai} T_{bj} H_{ab}^T \\
&= \sum_a T_{ai} (H^T T)_{aj} \\
&= (T^\dagger H^T T)_{ij} \\
F_{ij}^T &= T^\dagger H^T T
\end{aligned}$$

where $\dagger$ denotes the transpose. Now it follow that the determinant is $\det(T)^2 \det(H^T)$.

b

This is by Euler's theorem on F_j which has degree $n - 1$.

c

Write the matrix

$$H = \begin{bmatrix} f_{xx} & f_{xy} & f_{xz} \\ f_{xy} & f_{yy} & f_{yz} \\ f_{xz} & f_{yz} & f_{zz} \end{bmatrix}$$

if we add x times the first row and y times the second row to the third row we get

$$\frac{2}{z} \begin{bmatrix} f_{xx} & f_{xy} & f_{xz} \\ f_{xy} & f_{yy} & f_{yz} \\ f_x & f_y & f_z \end{bmatrix}$$

now doing the same to the columns we get:

$$\frac{2}{z^2} \begin{bmatrix} f_{xx} & f_{xy} & f_x \\ f_{xy} & f_{yy} & f_y \\ f_x & f_y & f \end{bmatrix}$$

We note that z is not zero on this point so it does not effect the intersection number. Then we can compute this determinant and aquire the result.

d

Note that if p is a multiple point of f then it lies on the varieties defined by the first derivaties. Now apply part c and property 5 of the intersection number.

e

Apply 3.12 part b to see that P is a flex if and only if $a = 0$. Furthermore note that p is an ordinary flex if and only if $d \neq 0$ by the proof we have given for 3.12.

Corollary

Use Bezout's theorem on H and F , therefore there is always atleast

5.25

Write $F = F_1 \cdots F_n$ where the F_i are irreducible of degree x_i and $G_i = F_1 \cdots F_i$ of degree $y_i = x_1 + \cdots + x_i$. We will prove the result by induction on n . The base case is just Theorem 2. We have $G_{k+1} = G_k F_{k+1}$. We want to count the sum of the multiplicities of the points. We first count separately, for G_k we have something smaller than:

$$\frac{(y_k - 1)(y_k - 2)}{2} + k - 1$$

and for F_{k+1} we have something smaller than:

$$\frac{(x_{k+1} - 1)(x_{k+1} - 2)}{2}$$

Now consider the points in which G_k and F_{k+1} both intersect. Note that there can be at most $y_k x_{k+1}$ such points. Moreover the additional contribution from one of these points will be

$$\frac{(m_1 + m_2)((m_1 + m_2) - 1)}{2} - \frac{m_1(m_1 - 1)}{2} - \frac{m_2(m_2 - 1)}{2} = m_1 m_2 \leq I(P, G_k \cap F_{k+1})$$

where m_1 and m_2 are the multiplicities at G_k and F_{k+1} respectively. Therefore the sum of these contributions to the sum is at most $y_k x_{k+1}$. Now write

$$\frac{(y_k + x_{k+1} - 1)(y_k + x_{k+1} - 2)}{2} - \frac{(y_k - 1)(y_k - 2)}{2} - \frac{(x_{k+1} - 1)(x_{k+1} - 2)}{2} = x_{k+1} y_k - 1$$

therefore the contribution is less than the change in $\frac{(n-1)(n-2)}{2} + c - 1$ when we add a simple component.

5.26

a

Make a change of coordinates to send P to $[0 : 1 : 0]$. If we plug in the line $L_\lambda = [\lambda : t : 1]$ then we evaluate F at $(\lambda : t : 1)$. This turns F into a polynomial in Y .

b

If L is a line intersecting this point we can write $L = aX + bZ$ and normalize to $L = X - \lambda^{-1}Z$ for some λ . Therefore the line has the desired form.

c

We must check that the leading order term does not vanish and that the polynomial is separable. These two condition both disqualify a finite number of choices of λ (since $F \cap F_Y$ is finite). Now we are done.

5.29

(1)

This is because the ring in question is a DVR. Furthermore H has order 1 since F and G meet transversally. Therefore H being contained in the maximal ideal is equivalent to it vanishing at P .

(2)

When P is a simple point on F we note that the intersection number and order of a curve (in the DVR) agree. Therefore the $AF + BG$ condition is euqivalent to the order of H being greater than or equal to the order of F .

(3)

Take

$$H = X^2 + Y^2 \quad F = X^2 \quad G = Y^2$$

and note that H has distinct tangents to F and G . Furthermore Noether's condition is clearly satisfied. However $m_P(H) \not\geq m_P(F) + m_P(G) - 1$.

5.30

Write $z = H/G$ and note that G and F must not intersect. Therefore Noether's condition is vacuously satisfied and we can write $H = AF + BG$. Therefore $z = AF/G + B \equiv B$, however B is a polynomial of degree 1 and must therefore be constant.

5.31

a

Note that as we vary the point near eachother the line between them becomes the tangent. If we draw a triangle inscribed on a conic then the intersections of the tangents with the opposite sides are collinear.

b

Pascals theorem says that the opposite sides intersect in collinear points. If we take a pentagon and replace one of the sides with a point and use the tangent line we get the analogues theorem.

c

Call the point P_1 . We will take the following steps.

- (a) Choose 4 other random points on the conic called $P_2, \dots, P_5$.
- (b) Construct the intersections $L_{12} \cap L_{54}$ and $L_{15} \cap L_{23}$.
- (c) These intersections define a line, R , take $Q = L \cap L_{34}$.
- (d) The line defined by Q and P_1 is tangent to P_1 at the conic.

5.32

Take the two cubics C and C' to be the opposite sides of the hexagon. Then we have $C \bullet C' = \sum_{i=1}^6 P_i + Q + R + S$ where Q, R, S are collinear (so there is L passing through them). Using $AF + BG$ on $H = C', F = C, G = L$ we obtain some B so that $H \bullet B = \sum_{i=1}^6 P_i$. By Bezout's thoerem this will be a conic which transcribes the hexagon.

5.33

Consider the cubic $R = L_1 L_2 L_3$. We note that $R \bullet C = \sum 2P_i + Q_i$. Now using characterization (3) of proposition 1 we can see that the conditions of $AF + BG$ are satisfied on $H = R, F = C, G = L^2$. Therefore we can find a B so that $B \bullet F = \sum Q_i$. Therefore B is the desired line.

5.34

Suppose that L is a line intersecting C on P_1, P_2, P_3 . let L_i be the tangent at P_i so that P_1 and P_2 are flexes. Suppose FTSOC that $I(P_3, L_3 \cap F) = 2$. We apply noether's theorem on $H = L_1 L_2 L_3, F = C, G = L^2$ to obtain B with $B \bullet C = P_1 + P_2 + Q$. Note that B is a line containing P_1 and P_2 so it must contain P_3 , a contradiction.

5.35

It suffices to show that the line operation is closed under nonsingular points. This is by Bezout's theorem.

5.36

Consider A, B , we will denote $\star$ as the line operation, $A \oplus_P B = P \star (A \star B)$ and $U = O \star O'$. Define the seven lines:

$$\begin{aligned} L_1 \bullet C &= P + U + \alpha(P) \\ L_2 \bullet C &= Q + U + \alpha(Q) \\ L_3 \bullet C &= O + (P \star Q) + (P \oplus Q) \\ L_4 \bullet C &= O' + (\alpha(P) \star \alpha(Q)) + (\alpha_P \oplus' \alpha(Q)) \\ L_5 \bullet C &= O + O' + U \\ L_6 \bullet C &= P + Q + (P \star Q) \\ L_7 \bullet C &= \alpha(P) + \alpha(Q) + (\alpha(P) \star \alpha(Q)) \end{aligned}$$

Now applying $AF + BG$ on $(L_1 + L_2 + L_3 + L_4 - L_5 - L_6 - L_7) \bullet C = (P \oplus Q) + U + (\alpha(P) \oplus' \alpha(Q))$ we obtain the line M so that $M \bullet C = (P \oplus Q) + U + (\alpha(P) \oplus' \alpha(Q))$. Therefore $\alpha(P \oplus Q) = \alpha(P) \oplus' \alpha(Q)$. This proves that it is a group homomorphism. It is an isomorphism because it is its own inverse.

5.37

a

Recall problem 5.23 and 5.34 which makes the following statements

- (a) A line passing through two flexes passes through another flex.
- (b) An irreducible cubic has 9 flexes.

which is enough to show that flexes are a subgroup of order 9. There are only two possible groups of order 9, those being C_9 and $C_3 \times C_3$. To show the latter it suffices to prove every nonzero element has order 3. Therefore consider the lines

$$\begin{aligned} L_1 &= P + P + (P \star P) = 3P \\ L_2 &= O + P + P \oplus P \\ L_3 &= P + P \oplus P + P \star (P \oplus P) \end{aligned}$$

Note that L_2 and L_3 share P and $P \otimes P$, therefore $P \star (P \oplus P) = O$ and taking $O \star$ on both sides $P \otimes P \otimes P = O$ since this point is a flex.

b

We need to show that any element of order 3 is a flex. Suppose that

$$P \oplus P \oplus P = O \star (P \star (O \star (P \star P))) = O$$

which simplifies to $P = P \star P$, the equation of a flex.

c

This is somewhat tautological.

d

The tangent line to a point is defined by $F_X(P)X + F_Y(P)Y + F_Z(P)Z = 0$. For this line to intersect $O = [0 : 1 : 0]$ we need $F_Y(P) = 0$ and $F(P) = 0$. Plugging this in and doing a bunch of algebra we get

$$[0 : 0 : 1] \quad [1 : 0 : 1] \quad [\lambda : 0 : 1]$$

e

WLOG make a projective change of coordinates so that $O = [0 : 1 : 0]$ and the tangent line to O is Z . Then by the result of 3.12 O is a flex iff:

$$F_* = ZG_* + aX^3 \quad G_* = 1 + \dots$$

As before we take the intersections of F_Y and F . We know from Bezout's that these curves will intersect 6 times. However, by considering the form of F_Y we know that it will be ZG_Y . Z intersects O , with multiplicity 3. Therefore G_Y will intersect C at 3 other points. None of these points will have intersection number larger than 1 since G_Y is a line which does not intersect O by construction and is therefore not tangent to any of them. Therefore there are exactly 3 other points. It now suffices to show that they are closed under the group operation. Consider A, B and O . Now write the lines

$$\begin{aligned} L_1 &= 2A + O \\ L_2 &= 2B + O \\ L_3 &= A + B + (A \star B) \\ L_4 &= O + (A \star B) + O \star (A \star B) \\ L_5 &= 3O \end{aligned}$$

and use $AF + BG$ on $(L_1 + L_2 + L_4 - L_3 - L_5) \bullet C = A + B + O \star (A \star B)$ to get a line through these points. Therefore $O \star (A \star B) = (A \star B)$ as desired.

f

We have already addressed the flex case in (e). Now as before make a change of coordinates so that $O = [0 : 1 : 0]$ and the tangent through O is Z . Therefore

$$F_* = ZG_* + aX^2 + bX^3 \quad G_* = 1 + \dots$$

and we attempt to compute the intersection of F with F_Y . We get

$$F_Y = Z(2Y + bZ + cX) + aX^2$$

whose tangent line at O is Z . Therefore the intersection number of F and F_Y at O is 2. We can calculate the tangent to F_Y at every point (A, B, C) as

$$(cC + 2aA)X + 2CY + (2bC + 2B + cA)Z$$

which will only have intersection number 2 if $C = 0$ since then the tangent will intersect O (and be the same as the tangent through the point). However by the form of F_Y , $C = 0$ implies that $A = 0$, therefore the only such point is O . Therefore there are 4 other points with order 2, which immediately implies they cannot form a group.

5.40

a

It suffices to show that the rational points are closed under the line operation. Let P and Q be the points and let $\bar{P}$ and $\bar{Q}$ be rational representatives in $\mathbb{A}^3$. We can parameterize the line as $r(t\bar{P} + (1-t)\bar{Q})$ for every $t, r \in k$. Then write $\bar{F}(t) = F(\bar{P}t + (1-t)\bar{Q})$. Note for a specific choice of scalars F will be a cubic in t with rational coefficients and two rational roots. Therefore the third root must be rational and we are done.

b

Since the rational points are closed under the line operation, they are closed under the group operation. Therefore this forms a subgroup.

Chapter 6

6.1

Suppose that a set U is open in Z . Then under the Y -induced topology it corresponds to U' in Y , with $U' \cap Z = U$. Then we have U'' with $U'' \cap Y = U'$. Since intersection is associative we have $U'' \cap Z = U$. Conversely if we have some $U'' \cap Z = U$ then we can take $U' = U'' \cap Z$ and $U' \cap Z = U$.

6.2

a

A set is closed in X if it is the intersection of a closed set and X . This immediately gives the forward direction. For the backward direction we have a collection of closed sets W_α so that $W_\alpha \cap U_\alpha = W \cap U_\alpha$. Now take R_α to be the complement of W_α . Note that $R_\alpha \cap U_\alpha$ is open since it is the intersection of two open sets. Now take the union of all the $R_\alpha \cap U_\alpha$ to obtain an open set R so that

$$R \cap U_\alpha = U_\alpha \setminus (U_\alpha \cap W)$$

and the complement W' of R is a closed set so that $X \cap W' = W$.

b

This is by (a) applied to the statement that a map is continuous iff the preimage of a closed set is closed.

6.3

a

We need to show that the preimage of a closed set is a closed set. The closed sets in $\mathbb{A}^1$ under the zariski topology are finite sets of points k_i . The preimage of this will be the union of the subvarieties generated by $f - k_i$.

b

The preimage of a polynomial map on an algebraic set is an algebraic set. Therefore the preimage of a closed set is a closed set.

6.4

a

It is quite clear that ϕ_i is bijective. Therefore we need to show that it and its inverse map closed sets to closed sets. This is by the correspondence established in the exposition in section 4.3:

"Except for projective varieties lying in H_∞ , there is a natural one-to-one correspondence between nonempty affine and projective varieties (see Problem 4.22)."

b

This is by the homomorphism property. For a set to be closed it suffices to check whether the image under the homomorphism is closed.

c

It is a property of homeomorphisms that they commute with taking the closure. This is by the definition of the closure as the intersection of all the closed sets containing a variety.

6.5

This theorem is actually only true for irreducible plane curves. Suppose that X is an infinite subset of V . We need to show that any open subset of V intersects X . This is equivalent to the statement that no closed subset of V contains X . However the curves contained in an irreducible variety are exactly the variety and finite sets of points.

Suppose that ϕ is a one-to-one mapping. We need to show that the preimage of a closed set is a closed set. The only closed sets of the image are the whole curve and finite sets of points. This corresponds to whole curve and finite sets of points in the domain. These are closed as desired. Furthermore the image of a finite set is a finite set of points therefore it maps closed sets to closed sets. If we assume that the map is a bijection then we are done.

6.6

It is clear that the condition is needed for continuity. Therefore it suffices to show that it is sufficient for continuity. Suppose that V is an algebraic set in $\mathbb{A}^n$. Due to the Noetherian condition we may write $\mathcal{I}(V) = (F_1, \dots, F_n)$ as finitely generated. Then V is the intersection of the hyperplanes defined by F_i . The preimage of an intersection is the intersection of the preimages so we have an intersection of closed sets, hence closed.

6.7

a

The fact that it is closed is by the definition of a closed subset of V . We need to show that $V(f)$ is not V unless f is 0. Suppose that $V(f)$ is V , then f is zero on all of V , and is therefore 0 in the quotient defined by $\Gamma(V)$.

b

This is true if V is a variety. Note that the subvarieties defined by f is an algebraic subset of V . By 6.5 this will be dense in V since it contains infinite points. Therefore it is the whole variety.

6.8

Write z as a fraction of coprime elements $z = a/b$. Note that $V \setminus U_z$ is the subvariety defined by a , therefore U_z is open. The mapping is continuous because the closed sets in k are finite sets of points $k_1, \dots, k_n$. The preimage of a point will be the ideal defined by $z - k_0$.